HR EMPLOYEE ATTRITION ANALYTICS PROJECT

Dataset Overview & Justification

Dataset Name	IBM HR Analytics Employee Attrition & Performance
Source	Kaggle (fictional dataset created by IBM data scientists)
Link	kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
Records	1,470
Columns	35 (base) / 47 (with engineered features)

Business Context

Employee attrition (voluntary/involuntary turnover) is one of the costliest problems in workforce management. Industry studies estimate replacing a single employee costs 50%–200% of their annual salary once recruiting, onboarding, lost productivity, and knowledge loss are accounted for. IBM compiled this dataset (with fictional but realistic employee records) to let analysts study the drivers of attrition across a large, multi-department organization spanning Sales, Research & Development, and Human Resources.

Business Problem

HR leadership needs to understand which employees are at the highest risk of leaving, why they leave, and which levers (compensation, work-life balance, career progression, overtime policy) the company can pull to retain high-value talent before attrition happens rather than reacting after the fact.

Analytical Objectives

1. Quantify attrition rate overall and by department, job role, and demographic segment.
2. Identify the strongest statistical drivers of attrition (satisfaction, income, overtime, tenure, commute distance, promotion history).
3. Build a predictive model that flags individual employees at elevated attrition risk.
4. Translate findings into KPIs and a monitoring dashboard for ongoing HR decision-making.

Why the Target Variable (Attrition) Is Suitable

Attrition is a binary Yes/No field recorded for every employee, giving a clean, unambiguous label for supervised classification. It is directly actionable (HR can intervene on a per-employee basis), commercially meaningful (turnover cost is a measurable P&L line), and it is influenced by a rich mix of numeric (income, tenure, satisfaction scores) and categorical (department, job role, marital status, overtime) predictors, making it ideal for both statistical EDA and machine learning.

Why This Dataset Is Appropriate

It satisfies every technical requirement of the brief: 1,470 records (>500), 35 columns (>8), a mix of ~15 categorical and ~20 numerical variables, embedded business metrics (MonthlyIncome, PercentSalaryHike, JobSatisfaction, PerformanceRating, YearsAtCompany), a clear ML target (Attrition), and rich correlational structure (e.g., OverTime, JobSatisfaction, and MonthlyIncome are all known to interact with Attrition).

Because the native file is unusually clean, realistic data-quality issues were deliberately injected into a working copy (see Data_RawIssues and Cleaning_Log tabs) to satisfy the data-cleaning requirements of the brief; all injected issues are fully documented and reversible against the original source data.

Expected Business Value

A working attrition-risk model lets HR prioritize retention spend (counter-offers, flexible work, promotion cycles) on the employees most likely to leave, rather than applying blanket policies. Even a modest reduction in voluntary attrition among high performers translates directly into avoided replacement cost, preserved institutional knowledge, and improved team morale.

Variable Categories Covered

Category	Example Variables
Demographic	Age, Gender, Marital Status
Compensation	MonthlyIncome, DailyRate, PercentSalaryHike, StockOptionLevel
Tenure	YearsAtCompany, TotalWorkingYears, YearsSinceLastPromotion
Satisfaction	JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance, RelationshipSatisfaction
Organizational	Department, JobRole, JobLevel, BusinessTravel
Target	Attrition (Yes / No)